

Transposition is the twist: the correction term $C_e^{(1)}$ as a height-reversal defect on level-one q -Fock space

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30 August 2026

Abstract

Let $\mathcal{F}$ be the level-one q -deformed Fock space of $U_q(\widehat{\mathfrak{sl}}_e)$, let P_e be the ribbon-sign operator $P_e|\lambda\rangle = \sum (-q)^h |\mu\rangle$ (sum over e -ribbon additions of height h), and let $B_{-1}^{[e]}$ be the Heisenberg mode $\sum (-q^{-1})^h$, so that $P_e = B_{-1}^{[e]} + (q - q^{-1})C_e^{(1)}$ (result C4). We prove that transposition $\Omega|\lambda\rangle = |\lambda'\rangle$ *reverses the ribbon height grading*, $\Omega N_e^{(h)} \Omega = N_e^{(e-1-h)}$, and deduce $\Omega P_e \Omega = (-q)^{e-1} B_{-1}^{[e]}$. Since $B_{-1}^{[e]}$ is the coefficientwise bar-conjugate of P_e , this identifies the abstract bar involution on this operator with a combinatorial involution, and answers Q55: $(q - q^{-1})C_e^{(1)}$ is exactly the deviation of P_e from its scalar-corrected transposed conjugate. **This last statement is not new:** it is the one-ribbon case of Leclerc–Thibon Prop. 7.10, which we locate and reconcile scalar-for-scalar in §3; what is new here is the graded form, and what follows. We then use the same circle of ideas to give a *structural* proof — independent of C4 — that $[e_i, P_e]$ is divisible by $(q - q^{-1})$, namely $[\bar{e}_i, P_e] = 0$; and we compute $\Omega e_i \Omega = q K_{-i}^{-1} \bar{e}_{-i}$ (again Leclerc–Thibon Prop. 7.10, rederived), which shows why Ω is not a crystal morphism. Sharpening this, we reduce the observed *unit monomial* form of the quotient, $[e_i, P_e]_{\nu\lambda} = \pm q^m (q - q^{-1})$, to a purely combinatorial statement about $[e_i, N_e^{(h)}]$ — all the q -arithmetic being forced by the single vanishing $[e_i, B_{-1}^{[e]}] = 0$. Finally we determine the i -signature word of the horizontal inflation $e\lambda$ to be $A^\delta(RA)^m$, which simultaneously (i) reproves $\varepsilon_i(e\lambda) \leq 1$ (C5 at level one), (ii) proves $\varepsilon_i = 0$ for the vertical inflation, thereby explaining the vertical/horizontal dichotomy as prefix-maximum versus suffix-maximum of one and the same ± 1 walk, and (iii) closes the gap in step (3b) of the C5 write-up identified by Lyra. **Everything here is level one.**

1 Setting and conventions

Fix $e \geq 2$. Let $\mathcal{F} = \bigoplus_\lambda \mathbb{Q}(q)|\lambda\rangle$, the sum over all partitions. We use English notation; the node in row r and column c (both 1-indexed) has *residue* $c - r \bmod e$. For a partition λ we set $\lambda_r = 0$ for $r > \ell(\lambda)$ and $\lambda_0 = +\infty$.

A *ribbon* (rim hook) is a finite skew shape μ/λ that is edge-connected and contains no 2×2 square. For an e -ribbon μ/λ we write $r(\mu/\lambda)$ for the number of rows it meets, $c(\mu/\lambda)$ for the number of columns, and

$$h(\mu/\lambda) := r(\mu/\lambda) - 1$$

for its *height* (equivalently spin/leg length; this convention is fixed throughout the programme and is the one implemented in `probes/2026-08-12-q34-jacon-lacabanne/qfock.py`).

Definition 1. For $0 \leq h \leq e - 1$ let $N_e^{(h)} \in \text{End}(\mathcal{F})$ be the $\mathbb{Q}(q)$ -linear *height- h ribbon counting operator*,

$$N_e^{(h)}|\lambda\rangle = \sum_{\substack{\mu \supset \lambda \\ \mu/\lambda \text{ an } e\text{-ribbon, } h(\mu/\lambda)=h}} |\mu\rangle.$$

For an indeterminate t put $R_e(t) := \sum_{h=0}^{e-1} t^h N_e^{(h)}$. Then

$$P_e = R_e(-q), \quad B_{-1}^{[e]} = R_e(-q^{-1}), \quad C_e^{(1)} = \sum_{h=0}^{e-1} (-1)^h [h]_q N_e^{(h)},$$

where $[h]_q = (q^h - q^{-h})/(q - q^{-1})$. Result C4 is the identity $P_e = B_{-1}^{[e]} + (q - q^{-1})C_e^{(1)}$, i.e. $(-q)^h - (-q^{-1})^h = (-1)^h (q - q^{-1})[h]_q$.

Let $b: \text{End}(\mathcal{F}) \rightarrow \text{End}(\mathcal{F})$ be the coefficientwise bar involution: apply $q \mapsto q^{-1}$ to every matrix entry in the basis $\{|\lambda\rangle\}$. It is a ring automorphism over $\mathbb{Z}$ (semilinear over $\mathbb{Z}[q^{\pm 1}]$). We write $\bar{X} = b(X)$. Since the $N_e^{(h)}$ have integer entries,

$$\overline{P_e} = B_{-1}^{[e]}, \quad \overline{C_e^{(1)}} = C_e^{(1)} \quad (1)$$

(the second because $[h]_q$ is bar-invariant).

Let $\Omega \in \text{End}(\mathcal{F})$ be transposition, $\Omega|\lambda\rangle = |\lambda'\rangle$, extended $\mathbb{Q}(q)$ -linearly; $\Omega^2 = \text{id}$. Let Ω_{bar} be the q -antilinear transposition, $\Omega_{\text{bar}}(\sum a_\lambda |\lambda\rangle) = \sum \overline{a_\lambda} |\lambda'\rangle$; also an involution. Write $t(X) := \Omega X \Omega$; then $\Omega_{\text{bar}} X \Omega_{\text{bar}} = t(\bar{X})$.

2 Transposition reverses the height grading

Lemma 2 (Ribbon rectangle). *If μ/λ is an e -ribbon then $r(\mu/\lambda) + c(\mu/\lambda) = e + 1$. Equivalently $c = e - h$ where $h = r - 1$.*

Proof. Let $S = \mu/\lambda$ and let r be the number of rows it meets. In each such row S occupies a nonempty contiguous interval of columns; relabel the rows of S as $1, \dots, r$ from top to bottom and let row i occupy $[a_i, b_i]$. Because λ and μ are partitions, $a_i \geq a_{i+1}$ and $b_i \geq b_{i+1}$. Edge-connectivity of S forces consecutive rows to share at least one column, i.e. $a_i \leq b_{i+1}$; absence of a 2×2 square forces them to share at most one, and their overlap is exactly $[a_i, b_{i+1}]$. Hence

$$b_{i+1} = a_i \quad (1 \leq i \leq r-1).$$

In particular the column intervals chain together, so the set of columns met by S is exactly $[a_r, b_1]$ and $c = b_1 - a_r + 1$. Now

$$e = \sum_{i=1}^r (b_i - a_i + 1) = \sum_{i=1}^r b_i - \sum_{i=1}^r a_i + r = \sum_{i=1}^r b_i - \left(\sum_{i=2}^r b_i + a_r \right) + r = b_1 - a_r + r = (c - 1) + r. \quad \square$$

Lemma 3 (Transposition of ribbon additions). *For partitions $\lambda \subset \mu$, the shape μ/λ is an e -ribbon if and only if μ'/λ' is, and in that case $r(\mu'/\lambda') = c(\mu/\lambda)$ and $c(\mu'/\lambda') = r(\mu/\lambda)$.*

Proof. The map $(r, c) \mapsto (c, r)$ is a bijection of $\mathbb{Z}_{\geq 0}^2$ carrying the Young diagram of ν to that of ν' for every partition ν ; it preserves inclusion, cardinality and edge-adjacency (hence connectivity), carries 2×2 squares to 2×2 squares, and interchanges rows with columns. $\square$

Theorem 4 (Height reversal). *For all $e \geq 2$ and $0 \leq h \leq e - 1$,*

$$\boxed{\Omega N_e^{(h)} \Omega = N_e^{(e-1-h)}}$$

Proof. Both sides are $\mathbb{Q}(q)$ -linear, so it suffices to evaluate on $|\lambda\rangle$. Since $\Omega^2 = \text{id}$,

$$\Omega N_e^{(h)} \Omega |\lambda\rangle = \Omega N_e^{(h)} |\lambda'\rangle = \Omega \sum_{\substack{\nu/\lambda' \text{ e-ribbon} \\ h(\nu/\lambda')=h}} |\nu\rangle = \sum_{\substack{\nu/\lambda' \text{ e-ribbon} \\ r(\nu/\lambda')=h+1}} |\nu'\rangle.$$

By Lemma 3 the assignment $\nu \mapsto \nu'$ is a bijection from $\{\nu : \nu/\lambda' \text{ an } e\text{-ribbon with } r = h + 1\}$ onto $\{\mu : \mu/\lambda \text{ an } e\text{-ribbon with } c = h + 1\}$, and by Lemma 2 the latter condition reads $r(\mu/\lambda) = e+1-(h+1) = e-h$, i.e. $h(\mu/\lambda) = e-1-h$. The right-hand sum is therefore exactly $N_e^{(e-1-h)} |\lambda\rangle$. $\square$

Corollary 5 (Functional equation). $\Omega R_e(t) \Omega = t^{e-1} R_e(t^{-1})$ as an identity of operators over $\mathbb{Z}[t^{\pm 1}]$.

Proof. $\Omega R_e(t) \Omega = \sum_h t^h N_e^{(e-1-h)} = \sum_k t^{e-1-k} N_e^{(k)} = t^{e-1} R_e(t^{-1})$. $\square$

Corollary 6 (The boxed identity of the brief).

$$\Omega P_e \Omega = (-q)^{e-1} B_{-1}^{[e]}, \quad \Omega B_{-1}^{[e]} \Omega = (-q^{-1})^{e-1} P_e.$$

Proof. Put $t = -q$ resp. $t = -q^{-1}$ in Corollary 5 and use $(-q)^{-1} = -q^{-1}$. $\square$

Corollary 7 (Transposition is bar, on this operator). $t(P_e) = (-q)^{e-1} b(P_e)$ and $t(B_{-1}^{[e]}) = (-q^{-1})^{e-1} b(B_{-1}^{[e]})$. Equivalently, with Ω_{bar} the antilinear transposition,

$$\Omega_{\text{bar}} P_e \Omega_{\text{bar}} = (-q^{-1})^{e-1} P_e,$$

so P_e is an eigenvector of the involution $X \mapsto \Omega_{\text{bar}} X \Omega_{\text{bar}}$ with eigenvalue $(-q^{-1})^{e-1}$.

Proof. Combine Corollary 6 with (1). $\square$

Corollary 8 (The spin normalisation trivialises the scalar). Work over $\mathcal{A} = \mathbb{Z}[t^{\pm 1/2}]$ with $t = -q$ and the involution $t \mapsto t^{-1}$ (which restricts to $q \mapsto q^{-1}$ on $\mathbb{Z}[q^{\pm 1}]$, since $t^{-1} = -q^{-1}$). Put $s = t^{(e-1)/2}$ and $\widehat{P}_e := s^{-1} P_e$. Then $\bar{s} = s^{-1}$ and

$$\Omega_{\text{bar}} \widehat{P}_e \Omega_{\text{bar}} = \widehat{P}_e.$$

Proof. $\Omega_{\text{bar}}(cX) \Omega_{\text{bar}} = \bar{c} \Omega_{\text{bar}} X \Omega_{\text{bar}}$ for a scalar c , so by Corollary 7, $\Omega_{\text{bar}} \widehat{P}_e \Omega_{\text{bar}} = \bar{s}^{-1} \cdot t^{-(e-1)} P_e = s s^{-2} P_e = s^{-1} P_e = \widehat{P}_e$, using $\bar{s} = t^{-(e-1)/2} = s^{-1}$ and $t^{-(e-1)} = (-q^{-1})^{e-1}$. $\square$

Remark 9. Corollary 8 is a Hilbert-90-shaped statement: the scalar $c = (-q^{-1})^{e-1}$ satisfies $c \bar{c} = 1$, and the “spin shift” by $(e-1)/2$ — the standard centring of the LLT spin statistic — is precisely a solution of $c = \bar{s}/s$. The half-integer power is unavoidable for e even, exactly as in LLT theory.

3 Relation to Leclerc–Thibon Proposition 7.10

Corollaries 6–8 are not new. They are the one-ribbon case of a published result, which I located on disk only after proving them, and I record the dictionary here rather than leave the attribution to a reader.

Lam ([math/0310250](#), [lam.tex:1026–1041](#), read directly) introduces the *semi-linear* involution $v \mapsto v'$ on $\mathcal{F}$ defined by $q' = q^{-1}$ and $|\lambda\rangle \mapsto |\lambda'\rangle$ — which is exactly our Ω_{bar} — and quotes:

Proposition 10 (Leclerc–Thibon, ASPM **28** (2000), Prop. 7.10; quoted at lam.tex:1030). *For all $u \in \mathcal{F}$ and compositions β with $|\beta| = k$,*

$$\begin{aligned}(e_i u)' &= q^{h-i-1} e_{-i} u', & (f_i u)' &= q^{-h-i-1} f_{-i} u', \\ (\mathcal{V}_\beta u)' &= (-q)^{(n-1)k} \tilde{\mathcal{V}}_\beta u', & (\mathcal{U}_\beta u)' &= (-q)^{(n-1)k} \tilde{\mathcal{U}}_\beta u',\end{aligned}$$

where $\mathcal{V}, \mathcal{U}$ are the horizontal and $\tilde{\mathcal{V}}, \tilde{\mathcal{U}}$ the vertical n -ribbon-strip operators.

The dictionary, checked rather than assumed:

- $n = e$.
- Lam’s spin of a single ribbon is its height, rows -1 (lam.tex:429–436: “the height of the ribbon r will then be called its spin $s(r)$ ”), i.e. our h .
- $\mathcal{V}_k|\lambda\rangle = \sum_\mu (-q)^{-s(\mu/\lambda)} |\mu\rangle$ (lam.tex:848). At $k = 1$ a horizontal ribbon strip of size 1 is a single e -ribbon and $(-q)^{-h} = (-q^{-1})^h$, so

$$\mathcal{V}_1 = B_{-1}^{[e]}.$$

A single ribbon is also a vertical strip of size 1, so $\tilde{\mathcal{V}}_1 = \mathcal{V}_1 = B_{-1}^{[e]}$ as well.

- Hence Prop. 10 at $k = 1$ reads $\Omega_{\text{bar}} B_{-1}^{[e]} \Omega_{\text{bar}} = (-q)^{e-1} B_{-1}^{[e]}$. Our Corollary 6 gives $\Omega_{\text{bar}} B_{-1}^{[e]} \Omega_{\text{bar}} = \overline{\Omega B_{-1}^{[e]}} \Omega = \Omega P_e \Omega = (-q)^{e-1} B_{-1}^{[e]}$. **The two agree, including the scalar.**
- Our Proposition 20 is the first two relations of Prop. 10: barring $\Omega e_i \Omega = q K_{-i}^{-1} \bar{e}_{-i}$ gives $\Omega_{\text{bar}} e_i \Omega_{\text{bar}} = q^{-1} K_{-i} e_{-i}$, which is $(e_i u)' = q^{h-i-1} e_{-i} u'$.

Two open questions from the 30 August reading log are therefore settled. *Q56 — is the linear Ω the same map as LT ’s semilinear involution?* No: they are different maps, related by $\Omega_{\text{bar}} = \text{b} \circ \Omega$, and both statements are true in their own form (Corollary 6 for Ω , Corollary 7 for Ω_{bar}); the passage between them is exactly $B_{-1}^{[e]} = \overline{P_e}$. *Q57 — does spin/cospin exchange account for the exponent?* Yes: Lemma 2 ($r + c = e + 1$) is the spin/cospin exchange for a single ribbon, and at $k = 1$ the exponent $(n - 1)k = e - 1$ matches on the nose.

What is not subsumed:

- Theorem 4 is a *graded* statement, q -free, at the level of the individual $N_e^{(h)}$; Prop. 10 is its q -specialised shadow at $k = 1$. (For general k the graded analogue is the statement that conjugation exchanges spin and cospin on ribbon tableaux, $s(T') = \text{mspin}(sh(T)) - s(T)$, which Lam also discusses.)
- Conversely Prop. 10 at $k \geq 2$ is *stronger* than anything proved here: it treats strips of several ribbons, where horizontal and vertical strips genuinely differ. We prove only $k = 1$.
- Theorem 15, Proposition 18, Conjecture 19, Theorem 25, Corollary 26 and the closing of the C5 step-(3b) gap are not consequences of Prop. 10 and are, as far as I know, new.

4 What transposition does to $C_e^{(1)}$

Proposition 11.

$$\Omega C_e^{(1)} \Omega = (-1)^{e-1} \sum_{k=0}^{e-1} (-1)^k [e-1-k]_q N_e^{(k)}.$$

In particular, for $e \geq 2$, $\Omega C_e^{(1)} \Omega$ is neither a scalar multiple of $C_e^{(1)}$ nor equal to $\overline{C_e^{(1)}}$.

Proof. Apply Theorem 4 termwise to $C_e^{(1)} = \sum_h (-1)^h [h]_q N_e^{(h)}$ and reindex $k = e-1-h$, using $(-1)^h = (-1)^{e-1-k}$. For the last claim take $e = 2$: $C_e^{(1)} = -N_e^{(1)}$ while $\Omega C_e^{(1)} \Omega = -N_e^{(0)}$, and $N_e^{(0)}, N_e^{(1)}$ are linearly independent (already on $|\emptyset\rangle$, where $N_e^{(0)}|\emptyset\rangle = |(2)\rangle$ and $N_e^{(1)}|\emptyset\rangle = |(1,1)\rangle$). Also $\overline{C_e^{(1)}} = C_e^{(1)}$ by (1). $\square$

So the correction term is *not* transposition-(anti)symmetric. Under Ω the quantum integer $[h]_q$ attached to height h is replaced by the *complementary* quantum integer $[e-1-h]_q$. This was one of the three candidates listed in the brief; the other two are refuted by Proposition 11 (and, computationally, by 381 explicit failures each — see §9).

5 The answer to Q55

Theorem 12. Define the $\mathbb{Z}[q^{\pm 1}]$ -linear map $\delta_\Omega: \text{End}(\mathcal{F}) \rightarrow \text{End}(\mathcal{F})$, $\delta_\Omega(X) := X - (-q)^{-(e-1)} \Omega X \Omega$. Then

$$\delta_\Omega(P_e) = P_e - B_{-1}^{[e]} = (q - q^{-1}) C_e^{(1)}.$$

Proof. Immediate from Corollary 6 and C4. $\square$

Thus $(q - q^{-1})C_e^{(1)}$ is a coboundary, for the following action and for no other reason: the order-two twisted conjugation $X \mapsto (-q)^{-(e-1)} \Omega X \Omega$ on $\text{End}(\mathcal{F})$, which is an honest involution after the spin renormalisation of Corollary 8 (before it, its square is multiplication by $(-q)^{-2(e-1)}$). We state it that way deliberately: this is not a statement of group cohomology.

Two further remarks, both of which limit the headline.

Corollary 13. $C_e^{(1)} = 0$ if and only if $e = 1$. In particular $C_e^{(1)}$ is not an artifact of a normalisation: no rescaling $P_e \mapsto uP_e$ makes it vanish.

Proof. $C_e^{(1)} = \sum_h (-1)^h [h]_q N_e^{(h)}$; the $N_e^{(h)}$ for $0 \leq h \leq e-1$ are nonzero and linearly independent (evaluate on $|\emptyset\rangle$, where the e -ribbon additions are exactly the hooks, one of each height), and $[h]_q \neq 0$ for $h \geq 1$. $\square$

Remark 14. What is an artifact of a convention is the scalar $(-q)^{e-1}$ in Corollary 6: it disappears in the spin-centred normalisation. The correction term itself measures a genuine reversal asymmetry, namely that $h \mapsto [h]_q$ is not symmetric under $h \mapsto e-1-h$. The brief's optimistic reading (“ $C_e^{(1)}$ demoted from mysterious correction to artifact of an inflation convention”) is therefore *too strong* and is not claimed here. What is achieved is the identification of the abstract bar involution $\bar{}$, on this operator, with the concrete combinatorial involution Ω .

6 A structural proof of the $(q - q^{-1})$ -divisibility of $[e_i, P_e]$

We now fix the Leclerc–Thibon action of $U_q(\widehat{\mathfrak{sl}}_e)$ on $\mathcal{F}$. For γ an addable or removable i -node of λ set

$$N_i^>(\lambda, \gamma) = \#\{\text{addable } i\text{-nodes of } \lambda \text{ strictly above } \gamma\} - \#\{\text{removable } i\text{-nodes strictly above}\},$$

and $N_i^<$ likewise with “below”. Then

$$f_i|\lambda\rangle = \sum_{\gamma \text{ addable } i} q^{N_i^>(\lambda, \gamma)} |\lambda + \gamma\rangle, \quad e_i|\lambda\rangle = \sum_{\gamma \text{ removable } i} q^{-N_i^<(\lambda - \gamma, \gamma)} |\lambda - \gamma\rangle,$$

and $K_i|\lambda\rangle = q^{\langle h_i, \text{wt } \lambda \rangle} |\lambda\rangle$ with $\langle h_i, \text{wt } \lambda \rangle = A_i(\lambda) - R_i(\lambda)$ (numbers of addable and removable i -nodes). We verified $[e_i, f_j] = \delta_{ij}(K_i - K_i^{-1})/(q - q^{-1})$ on 665 instances as an independent witness that this is the genuine module (§9).

We take as given (Uglov; used already in C4) the level-one commutation

$$[e_i, B_{-1}^{[e]}] = [f_i, B_{-1}^{[e]}] = 0 \quad (i \in \mathbb{Z}/e). \quad (2)$$

Theorem 15. *Let $\bar{e}_i = b(e_i)$, $\bar{f}_i = b(f_i)$ and put*

$$E_i := \frac{e_i - \bar{e}_i}{(q - q^{-1})}, \quad F_i := \frac{f_i - \bar{f}_i}{(q - q^{-1})}.$$

Then E_i, F_i have entries in $\mathbb{Z}[q^{\pm 1}]$ — explicitly, the entry of E_i at $(\lambda - \gamma, \lambda)$ is $-[N_i^<]_q$ and that of F_i at $(\lambda + \gamma, \lambda)$ is $[N_i^>]_q$ — and

$$[\bar{e}_i, P_e] = [\bar{f}_i, P_e] = 0, \quad [e_i, P_e] = (q - q^{-1})[E_i, P_e], \quad [f_i, P_e] = (q - q^{-1})[F_i, P_e].$$

In particular every entry of $[e_i, P_e]$ and of $[f_i, P_e]$ is divisible by $(q - q^{-1})$ in $\mathbb{Z}[q^{\pm 1}]$. (This is Conjecture 4 of 2026-08-12-commutation-defect.tex and the statement the registry calls “T2”; see Remark 17 for the label collision.)

Proof. b is a ring automorphism of $\text{End}(\mathcal{F})$ over $\mathbb{Z}$, so applying it to (2) and using $b(B_{-1}^{[e]}) = P_e$ from (1) gives $[\bar{e}_i, P_e] = [\bar{f}_i, P_e] = 0$. Subtracting this from $[e_i, P_e]$ gives $[e_i, P_e] = [e_i - \bar{e}_i, P_e]$. Finally $e_i - \bar{e}_i$ is entrywise $q^{-N} - q^N = -(q - q^{-1})[N]_q$, hence divisible by $(q - q^{-1})$ with quotient in $\mathbb{Z}[q^{\pm 1}]$; likewise $f_i - \bar{f}_i$ is entrywise $q^N - q^{-N} = (q - q^{-1})[N]_q$. (Here $[-N]_q = -[N]_q$, so negative N is covered.) $\square$

This is the structural reason the brief asked for, and it is *independent of C4*: it does not use the decomposition $P_e = B_{-1}^{[e]} + (q - q^{-1})C_e^{(1)}$, only (2) and the fact that $P_e = \overline{B_{-1}^{[e]}}$. The slogan is: $B_{-1}^{[e]}$ commutes with the Chevalley operators, so P_e commutes with their bar-conjugates, and the two differ by $(q - q^{-1})$ times an integral operator. Comparing with the C4 route gives a nontrivial identity.

Corollary 16. $[e_i, C_e^{(1)}] = [E_i, P_e]$ and $[f_i, C_e^{(1)}] = [F_i, P_e]$.

Proof. By C4 and (2), $[e_i, P_e] = (q - q^{-1})[e_i, C_e^{(1)}]$; compare with Theorem 15 and cancel $(q - q^{-1})$ (which is a nonzerodivisor). $\square$

Remark 17 (A label collision, resolved). The names “T2/T3” are used differently in two places in this programme, and the difference matters. In `2026-08-12-commutation-defect.tex`, T2 is the $q = 1$ statement $[e_i, P_e]_{q=1} = 0$ and T3 is the $(q - 1)$ -divisibility of $e_i \cdot v_{k', e}$; the $(q - q^{-1})$ -divisibility is *Conjecture 4* there, explicitly flagged as empirical. In the registry node `T2-qdiff-divisibility` and in the 30 August brief, “T2” names the $(q - q^{-1})$ -divisibility of $[e_i, P_e]$ and “T3” the claim that the quotient is linear in q . Theorem 15 above proves the *mathematical* statement — $(q - q^{-1})$ -divisibility of every entry of $[e_i, P_e]$ and $[f_i, P_e]$ — which was the 2026-08-12 Conjecture 4 and is the registry’s “T2”. It does not touch the 2026-08-12 T2/T3, which are proved there already. No result is affected; the labels are.

6.1 A sharper divisibility, and a clean reduction

The registry’s “T3” — that the quotient is linear in q — is not what the data say. The data say something much stronger and much simpler.

Proposition 18. *Fix $e \geq 2$, $i \in \mathbb{Z}/e$, $x \in \{e_i, f_i\}$ and partitions λ, ν . Put $c_h := ([x, N_e^{(h)}])_{\nu\lambda} \in \mathbb{Z}[q^{\pm 1}]$ and $H := \{h : c_h \neq 0\}$. Then:*

1. $|H| \neq 1$.
2. Suppose $|H| = 2$, say $H = \{b, b + d\}$ with $d \geq 1$, and suppose $c_b = \varepsilon q^k$ with $\varepsilon \in \{\pm 1\}$. Then necessarily $c_{b+d} = -\varepsilon(-1)^d q^{k+d}$, and

$$([x, P_e])_{\nu\lambda} = -\varepsilon(-1)^b q^{k+b+d} (q - q^{-1}) [d]_q.$$

In particular the quotient by $(q - q^{-1})$ is a unit monomial $\pm q^m$ if and only if $d = 1$.

Proof. Let $\Phi(t) := \sum_{h=0}^{e-1} c_h t^h \in \mathbb{Z}[q^{\pm 1}][t]$. Since $R_e(t)$ is $\mathbb{Z}[t]$ -linear in the $N_e^{(h)}$, we have $\Phi(t) = ([x, R_e(t)])_{\nu\lambda}$; hence

$$\Phi(-q^{-1}) = ([x, B_{-1}^{[e]}])_{\nu\lambda} = 0 \quad \text{and} \quad \Phi(-q) = ([x, P_e])_{\nu\lambda},$$

the first by (2).

- (1) If $H = \{b\}$ then $0 = \Phi(-q^{-1}) = c_b(-q^{-1})^b$, so $c_b = 0$, a contradiction.
- (2) From $c_{b+d}(-q^{-1})^{b+d} + c_b(-q^{-1})^b = 0$ we get $c_{b+d} = -c_b(-1)^d q^d = -\varepsilon(-1)^d q^{k+d}$. Then

$$\Phi(-q) = c_{b+d}(-q)^{b+d} + c_b(-q)^b = \varepsilon(-1)^b \left[-q^{k+d+b+d} + q^{k+b} \right] = -\varepsilon(-1)^b q^{k+b} (q^{2d} - 1),$$

using $(-1)^d(-1)^{b+d} = (-1)^b$. Finally $q^{2d} - 1 = q^d(q^d - q^{-d}) = q^d(q - q^{-1})[d]_q$. $\square$

So *all* the q -arithmetic is forced by the single vanishing $\Phi(-q^{-1}) = 0$. What remains is purely combinatorial.

Conjecture 19. For all $e \geq 2$, $i \in \mathbb{Z}/e$, $x \in \{e_i, f_i\}$ and all λ, ν :

- (P2) every entry $([x, N_e^{(h)}])_{\nu\lambda}$ is either 0 or a unit monomial $\pm q^k$; and
- (P1) the support H is either empty or a pair of *consecutive* integers.

By Proposition 18 these imply that every entry of $[e_i, P_e]$ and of $[f_i, P_e]$ is exactly $\pm q^m(q - q^{-1})$ for some $m \in \mathbb{Z}$ — a unit monomial quotient, far stronger than linearity in q .

Evidence, over $e \leq 6$ and $|\lambda| \leq 10$: (P2) holds for all 5366 nonzero entries of the operators $[x, N_e^{(h)}]$ tested; (P1) holds for all 2683 nonempty supports, every one of which has $|H| = 2$ with the two indices consecutive; and, computed directly and independently of the reduction, all 2683 nonzero entries of $[e_i, P_e]$ and $[f_i, P_e]$ are of the form $\pm q^m(q - q^{-1})$. Zero exceptions in each case.

Note that Proposition 18(1) already rules out $|H| = 1$ unconditionally, so the content of (P1) is the bound $|H| \leq 2$ together with consecutiveness; and $|H| = 0$ is exactly the case of a vanishing commutator entry. Theorem 15 remains the only *proved* divisibility statement here; Conjecture 19 is the sharp form, and the combinatorial statement it reduces to — an explicit commutation rule between a Chevalley operator and ribbon addition at fixed height — looks to me like the right next target, replacing the ill-posed “T3 linearity”.

7 Transposition on the Chevalley generators

Proposition 20. *For all $i \in \mathbb{Z}/e$,*

$$\Omega e_i \Omega = q K_{-i}^{-1} \bar{e}_{-i}, \quad \Omega f_i \Omega = \bar{f}_{-i} \cdot q^{-1} K_{-i},$$

where in the first identity K_{-i}^{-1} acts on the target and in the second K_{-i} acts on the source.

Proof. Transposition sends the node (r, c) to (c, r) , hence sends residue i to $-i$, addable nodes to addable nodes, removable to removable, and the relation “strictly above” to “strictly to the left”. By Lemma 22 below, within the set of addable and removable i -nodes of a partition, “strictly to the left” coincides with “strictly below”. Hence $N_i^<(\lambda, \gamma) = N_{-i}^>(\lambda', \gamma')$ and $N_i^>(\lambda, \gamma) = N_{-i}^<(\lambda', \gamma')$. Therefore $\Omega e_i \Omega$ removes $(-i)$ -nodes with coefficient $q^{-N_{-i}^>}$ and $\Omega f_i \Omega$ adds $(-i)$ -nodes with coefficient $q^{N_{-i}^<}$. Now for a node γ addable to ν ,

$$N_j^>(\nu, \gamma) + N_j^<(\nu, \gamma) = (A_j(\nu) - 1) - R_j(\nu) = \langle h_j, \text{wt } \nu \rangle - 1,$$

since γ itself is one of the $A_j(\nu)$ addable j -nodes. Applying this with $\nu = \lambda - \gamma$ (for e_i) and $\nu = \lambda$ (for f_i) converts $q^{-N^>}$ into $q^{1-\langle h \rangle} q^{N^<}$ and $q^{N^<}$ into $q^{\langle h \rangle - 1} q^{-N^>}$, which are the two stated formulas. $\square$

Corollary 21. $[\Omega e_i \Omega, P_e] = [\Omega f_i \Omega, P_e] = 0$.

Proof. Conjugate (2) by Ω and use Corollary 6: $0 = \Omega[e_i, B_{-1}^{[e]}] \Omega = [\Omega e_i \Omega, (-q^{-1})^{e-1} P_e]$. $\square$

So P_e commutes with two visibly different deformations of the Chevalley action — the bar-conjugate one and the transposed one — and Proposition 20 says they differ only by the diagonal operator $q^{\pm 1} K_{\mp i}^{\pm 1}$, with which P_e commutes because an e -ribbon contains exactly one node of each residue and so preserves $\langle h_j, \text{wt } \cdot \rangle$. The two statements are one statement.

8 The crystal side: why Ω is not a crystal morphism

Lemma 22 (Order reversal). *Let $e \geq 2$, λ a partition, $i \in \mathbb{Z}/e$, and let $\mathcal{N}_i(\lambda)$ be the set of addable and removable i -nodes of λ . If $(r_1, c_1), (r_2, c_2) \in \mathcal{N}_i(\lambda)$ with $r_1 < r_2$ then $c_1 > c_2$.*

Proof. First note that every $(r, c) \in \mathcal{N}_i(\lambda)$ satisfies

$$\lambda_r \leq c \leq \lambda_r + 1, \quad \lambda_{r-1} \geq c, \quad \lambda_{r+1} \leq c - 1. \quad (3)$$

Indeed a removable node is (r, λ_r) with $\lambda_{r+1} < \lambda_r$, and an addable node is $(r, \lambda_r + 1)$ with $\lambda_{r-1} > \lambda_r$, i.e. $\lambda_{r-1} \geq \lambda_r + 1$.

Now take $r_1 < r_2$. If $r_2 \geq r_1 + 2$ then $\lambda_{r_2} \leq \lambda_{r_1+1} \leq c_1 - 1$ by (3), so $c_2 \leq \lambda_{r_2} + 1 \leq c_1$. If equality held we would have $c_2 = c_1$ and $\lambda_{r_2} = c_2 - 1$, so (r_2, c_2) is addable and $\lambda_{r_2-1} \geq c_2 = c_1$; but $r_2 - 1 \geq r_1 + 1$ gives $\lambda_{r_2-1} \leq \lambda_{r_1+1} \leq c_1 - 1$, a contradiction. Hence $c_1 > c_2$.

If $r_2 = r_1 + 1$ then again $\lambda_{r_2} \leq c_1 - 1$ and $c_2 \leq c_1$, and equality forces $\lambda_{r_2} = c_1 - 1$, (r_2, c_2) addable, and $\lambda_{r_1} = \lambda_{r_2-1} \geq c_1$; combined with $\lambda_{r_1} \leq c_1$ this gives $\lambda_{r_1} = c_1$, so (r_1, c_1) is removable. The two residues are then $c_1 - r_1$ and $c_1 - r_1 - 1$, which are distinct modulo e because $e \geq 2$ — contradicting that both nodes have residue i . Hence $c_1 > c_2$. $\square$

Definition 23. The i -signature word $w_i(\lambda) \in \{A, R\}^*$ is obtained by listing the elements of $\mathcal{N}_i(\lambda)$ in order of increasing row and recording A for addable, R for removable. For $w \in \{A, R\}^*$ let $\text{exc}(u) = \#R(u) - \#A(u)$ and set

$$\varepsilon(w) = \max_{u \text{ prefix of } w} \text{exc}(u), \quad \varepsilon^{\text{op}}(w) = \max_{u \text{ suffix of } w} \text{exc}(u)$$

(both maxima include the empty word, hence are ≥ 0). Then $\varepsilon_i(\lambda) = \varepsilon(w_i(\lambda))$ is the Kashiwara ε of the level-one Fock crystal: it is the number of R 's surviving repeated cancellation of factors AR . Dually ε^{op} counts the R 's surviving cancellation of factors RA .

Theorem 24. For every partition λ and every i ,

$$w_{-i}(\lambda') = \overleftarrow{w_i(\lambda)} \quad \text{and hence} \quad \varepsilon_{-i}(\lambda') = \varepsilon^{\text{op}}(w_i(\lambda)).$$

Proof. Transposition is a bijection $\mathcal{N}_i(\lambda) \rightarrow \mathcal{N}_{-i}(\lambda')$ preserving the type (A or R) and exchanging rows with columns. By Lemma 22 the row order on $\mathcal{N}_i(\lambda)$ is the reverse of the column order, so reading $\mathcal{N}_{-i}(\lambda')$ by increasing row means reading $\mathcal{N}_i(\lambda)$ by increasing column, i.e. by decreasing row. For the second statement, reversing a word exchanges prefixes with suffixes. $\square$

Since $\varepsilon \neq \varepsilon^{\text{op}}$ in general (already on $w = RA$, where $\varepsilon = 1$ and $\varepsilon^{\text{op}} = 0$), Ω is not a morphism of $\widehat{\mathfrak{sl}}_e$ -crystals; we recorded 490 explicit instances of $\varepsilon_{-i}(\lambda') \neq \varepsilon_i(\lambda)$ in the range tested. This is consistent with, and explained by, Proposition 20: $\Omega e_i \Omega$ is not e_{-i} but its bar-conjugate twisted by a diagonal, and bar exchanges the lower crystal lattice with the upper one.

8.1 The signature word of a horizontal inflation

Write $e\lambda := (e\lambda_1, e\lambda_2, \dots)$ for the *horizontal* inflation, and $V(\lambda)$ for the *vertical* inflation, in which every part of λ is repeated e times. These are exchanged by transposition:

$$V(\lambda) = (e\lambda')'. \quad (4)$$

(The transpose of $e\mu$ has each part of μ' repeated e times.)

Theorem 25. Let λ be a partition, $\ell = \ell(\lambda)$, $\mu = e\lambda$, and $i \in \mathbb{Z}/e$. Then

$$w_i(\mu) = A^\delta (RA)^m, \quad \delta = [i \equiv 0], \quad m = \#\{1 \leq r \leq \ell : r \equiv -i \pmod{e}, \lambda_r > \lambda_{r+1}\}.$$

Proof. Since $\mu_r = e\lambda_r \equiv 0 \pmod{e}$:

- the removable nodes of μ are the $(r, e\lambda_r)$ with $1 \leq r \leq \ell$ and $\lambda_r > \lambda_{r+1}$; such a node has residue $e\lambda_r - r \equiv -r$;

- the addable nodes of μ are the $(s, e\lambda_s + 1)$ with $1 \leq s \leq \ell + 1$ and $\lambda_{s-1} > \lambda_s$ (recall $\lambda_0 = +\infty$, $\lambda_{\ell+1} = 0$); such a node has residue $e\lambda_s + 1 - s \equiv 1 - s$.

Substituting $s = r + 1$ turns the addable condition into $\lambda_r > \lambda_{r+1}$ and the addable residue into $-r$. Hence:

$$\boxed{\begin{array}{ll} \mu \text{ has a removable node in row } r & \iff \lambda_r > \lambda_{r+1}, \\ \mu \text{ has an addable node in row } r + 1 & \iff \lambda_r > \lambda_{r+1}, \end{array}} \quad (5)$$

and when they exist both have residue $-r \bmod e$. Consequently $\mathcal{N}_i(\mu)$ is the disjoint union of

- the pairs $\{(r, e\lambda_r), (r + 1, e\lambda_{r+1} + 1)\}$ over $1 \leq r \leq \ell$ with $r \equiv -i$ and $\lambda_r > \lambda_{r+1}$; and
- the single addable node $(1, 1)$, present iff $1 \equiv 1 - i$, i.e. iff $i \equiv 0$ (this is the only addable node whose partner index $r = 0$ lies outside $[1, \ell]$; note $\lambda_0 > \lambda_1$ holds by convention).

Each pair occupies the consecutive rows $r, r + 1$ and contributes RA in reading order. Distinct pairs have starting rows differing by a multiple of $e \geq 2$, so they occupy disjoint and non-interleaving row blocks $\{r, r + 1\}, \{r + e, r + e + 1\}, \dots$. The lone addable node, when present, is in row 1, while pairs then start at rows $\equiv 0 \bmod e$, i.e. at $r \geq e \geq 2$; so it precedes them all. This is exactly $A^\delta(RA)^m$. $\square$

Corollary 26. *For all λ and all i :*

1. $\varepsilon_i(e\lambda) \leq 1$, with equality iff $i \not\equiv 0$ and $m \geq 1$; in particular $\varepsilon_0(e\lambda) = 0$ always. (This is C5 at level one, reproved.)
2. $\varepsilon^{\text{op}}(w_i(e\lambda)) = 0$.
3. $\varepsilon_i(V(\lambda)) = 0$.

Proof. By Theorem 25 the word is $A^\delta(RA)^m$. Its prefixes have excess ≤ 1 , with 1 attained iff the word begins with R , i.e. $\delta = 0$ and $m \geq 1$; this gives (1). Its suffixes all end in A (if nonempty) and have excess ≤ 0 — the suffix $(RA)^k$ has excess 0 and $A(RA)^k$ has excess -1 — which gives (2). For (3), combine (4) with Theorem 24: $\varepsilon_i(V(\lambda)) = \varepsilon_i((e\lambda)') = \varepsilon^{\text{op}}(w_{-i}(e\lambda')) = 0$. $\square$

This is the promised reconciliation. *The vertical/horizontal dichotomy is prefix-maximum versus suffix-maximum of one and the same ± 1 walk.* The word $RARA \cdots RA$ has a lone uncanceled R at its front and none at its back; transposition reverses it; the operator statement (Corollary 6) and the crystal statement (Corollary 26) are two readings of the same involution Ω .

8.2 A gap in the C5 write-up, closed

Lyra's review of 2026-08-11-C5-gerber-bicrystal.tex (30 August 2026) observed that step (3b) asserts, without proof, that the addable node of row r and the removable node of row $r - 1$ both exist precisely when $\lambda_{r-1} > \lambda_r$, and identified this coincidence of booleans as the load-bearing fact. Display (5) in the proof of Theorem 25 is exactly that statement, and its proof is the single observation that every row of $e\lambda$ has length divisible by e : the corner of row r then sits in a column $\equiv 0$, giving residue $-r$, while the addable node of row $r + 1$ sits in a column $\equiv 1$, giving residue $-r$ as well; and both validity conditions unwind to $\lambda_r > \lambda_{r+1}$ because inflation by e is strictly increasing on parts. The gap is closed.

9 Computational verification

All probes are in `probes/2026-08-30-Q55-transpose/`. Ribbon additions were computed by two independent routes — an abacus/ β -number route (the implementation used throughout this programme) and a brute-force Young-diagram route that enumerates all $\mu \supset \lambda$ with μ/λ connected, of size e , and free of 2×2 squares. The two agree on all 239 pairs (e, λ) tested, which pins the height convention $h = \text{rows} - 1$ in both.

Statement	Range	Checks	Failures
route A = route B (ribbon additions)	$e \leq 6, \lambda \leq 8$	239	0
Lemma 2 $r + c = e + 1$	$e \leq 6, \lambda \leq 8$	1052	0
Thm 4 (partition level)	$e \leq 8, \lambda \leq 14$	3556 (21910 additions)	0
Thm 4 (operator level, per h)	$e \leq 7, \lambda \leq 9$	1458	0
Cor. 6 $\Omega P_e \Omega = (-q)^{e-1} B_{-1}^{[e]}$	$e \leq 7, \lambda \leq 9$	381	0
Cor. 6 $\Omega B_{-1}^{[e]} \Omega = (-q^{-1})^{e-1} P_e$	$e \leq 7, \lambda \leq 9$	381	0
Cor. 7 $\Omega_{\text{bar}} P_e \Omega_{\text{bar}} = (-q^{-1})^{e-1} P_e$	$e \leq 7, \lambda \leq 9$	381	0
Prop. 11 (complementary form)	$e \leq 7, \lambda \leq 9$	381	0
<i>refuted</i> : $\Omega C_e^{(1)} \Omega = -(-q)^{e-1} C_e^{(1)}$	$e \leq 7, \lambda \leq 9$	381	381
<i>refuted</i> : $\Omega C_e^{(1)} \Omega = \overline{C_e^{(1)}}$	$e \leq 7, \lambda \leq 9$	381	381
$[e_i, f_i] = (K_i - K_i^{-1})/(q - q^{-1})$	$e \leq 5, \lambda \leq 8$	665	0
(2) $[e_i, B_{-1}^{[e]}] = [f_i, B_{-1}^{[e]}] = 0$	$e \leq 5, \lambda \leq 8$	2×665	0
Thm 15 $[\bar{e}_i, P_e] = [\bar{f}_i, P_e] = 0$	$e \leq 5, \lambda \leq 8$	2×665	0
Thm 15 $[e_i, P_e] = (q - q^{-1})[E_i, P_e]$	$e \leq 5, \lambda \leq 8$	665	0
Thm 15 $[f_i, P_e] = (q - q^{-1})[F_i, P_e]$	$e \leq 5, \lambda \leq 8$	665	0
Cor. 16 $[e_i, C_e^{(1)}] = [E_i, P_e]$	$e \leq 5, \lambda \leq 8$	665	0
Cor. 16 $[f_i, C_e^{(1)}] = [F_i, P_e]$	$e \leq 5, \lambda \leq 8$	665	0
Prop. 20 (both forms, e and f)	$e \leq 5, \lambda \leq 8$	4×665	0
Lemma 22 (columns decrease)	$e \leq 6, \lambda \leq 10$	2780	0
Thm 24 (word reversal)	$e \leq 6, \lambda \leq 10$	2780	0
Thm 24 (ε form)	$e \leq 6, \lambda \leq 10$	2780	0
(4) $V(\lambda) = (e\lambda)'$	$e \leq 6, \lambda \leq 8$	335	0
Cor. 26(1) $\varepsilon_i(e\lambda) \leq 1$	$e \leq 6, \lambda \leq 8$	1340	0
Cor. 26(2) $\varepsilon^{\text{op}} = 0$	$e \leq 6, \lambda \leq 8$	1340	0
Cor. 26(3) $\varepsilon_i(V(\lambda)) = 0$	$e \leq 6, \lambda \leq 8$	1340	0
(5) the S_r coupling (both halves)	$e \leq 7, \lambda \leq 10$	2×4890	0
(5) residues both $\equiv -r$	$e \leq 7, \lambda \leq 10$	2×1704	0
Thm 25 $w_i(e\lambda) = A^\delta(RA)^m$	$e \leq 7, \lambda \leq 10$	3753	0
Conj. 19(P2) unit monomial entries	$e \leq 6, \lambda \leq 10$	5366	0
Conj. 19(P1) $ H = 2$, consecutive	$e \leq 6, \lambda \leq 10$	2683	0
$[x_i, P_e]_{\nu\lambda} = \pm q^m(q - q^{-1})$ (direct)	$e \leq 6, \lambda \leq 10$	2683	0

Non-vacuity: in the range $e \leq 6, |\lambda| \leq 10$ there are 490 instances with $\varepsilon_{-i}(\lambda') \neq \varepsilon_i(\lambda)$, so Theorem 24 is not asserting an equality that holds for trivial reasons. The two *refuted* rows above are the other two candidates the brief asked to be tested; they fail on every instance where the commutator is nonzero, the smallest being $e = 2, \lambda = \emptyset$.

10 What is not closed

1. **The sharp divisibility is not proved.** Theorem 15 gives $(q - q^{-1})$ -divisibility structurally, but not the observed *unit monomial* quotient. Proposition 18 reduces that to the purely combinatorial Conjecture 19 ((P1)+(P2)), which is the natural next target. Note that the route through E_i in Theorem 15 cannot see it: the entries of E_i are quantum integers $[N_i^<]_q$ of unbounded degree, so the monomial shape must arise from cancellation inside $[E_i, P_e]$, which that argument does not control. The registry’s phrasing “the quotient is linear in q ” should be retired in favour of the monomial statement (see Remark 17).
2. **Level ≥ 2 .** Everything above is level one. Lemma 2 and Theorem 4 are statements about single ribbons and should lift verbatim to each component of a multipartition; the identification of Ω with the vertical/horizontal exchange, and Theorem 25, will *not* lift unchanged, because the level- ℓ signature is a merge by content of the per-component words and transposition must also act on the multicharge $s \mapsto -s$ (reversed). Nothing here resolves the $\ell \geq 2$ obstruction recorded at `C5-level-ell-individual-bound`, and nothing here needs Uglov Prop. 3.16.
3. **Ω_{bar} and the canonical basis — partly answered, in the literature.** Ω_{bar} is exactly the semilinear involution of Prop. 10 (§3), and Lam draws from that proposition the immediate consequence $(G_{n\lambda})' = (-q)^{(n-1)k} G_{n\lambda'}$: so Ω_{bar} *does* permute the canonical basis elements indexed by inflations, up to the same scalar. What is *not* settled is the behaviour on G_μ for general μ , and hence whether this is a genuine transposition symmetry of the LLT polynomials rather than a statement about the e -inflated locus. That is the question I would take next on this branch, and it should be attacked by reading LT ASPM 28 §7 in the original rather than through Lam’s citation.
4. **The “artifact” reading is refuted.** Corollary 13 shows $C_e^{(1)} \neq 0$ for every $e \geq 2$ in every normalisation. The brief’s hoped-for demotion of $C_e^{(1)}$ to a convention artifact does not survive; what survives is the identification of the bar involution with transposition on this operator.
5. **Rick’s β' programme.** Nothing here is a claim about Rick’s objects. The present results are consistent with the suspicion that his “vertical versus connected” obstruction and the inflation-convention obstruction are the same one — Corollary 26 makes that precise *at level one*, for ε_i , and no further — but the identification of his κ_μ with P_2 has not been checked and is not asserted.